Cryptography & Network Security Essentials

Course: Cryptography | Semester 8 | Contributor: Rohan Verma

Symmetric ciphers AES, Asymmetric crypto RSA, ECC, Diffie-Hellman key exchange, and SHA-256.

Symmetric & Asymmetric Cryptography

* AES block cipher round structure: SubBytes, ShiftRows, MixColumns, AddRoundKey.
* RSA algorithm: prime generation, Euler totient phin, modular exponentiation, and decryption.
* Diffie-Hellman key exchange protocol and vulnerability to Man-In-The-Middle without authentication.

Cryptographic Hashes & Digital Signatures

* SHA-256 properties: pre-image resistance, second pre-image resistance, collision resistance.
* Digital Signature Algorithm DSA providing integrity, authenticity, and non-repudiation.
* Public Key Infrastructure PKI, X.509 certificate authorities, and TLS 1.3 handshake.

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